

## Presentation 5

- Links to presentation(s) and code(s) on GitHub
  - Presentation (just my slides – see Marcos' slides for full slideshow):
    - [https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Presentations/Verbiage%20analysis/CallAbandonment\\_08Dec\\_Adam.pdf](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Verbiage%20analysis/CallAbandonment_08Dec_Adam.pdf)
  - Code:  
[https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Code/Verbiage%20analysis/ABANDONMENT\\_final\\_08Dec\\_Adam.ipynb](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/ABANDONMENT_final_08Dec_Adam.ipynb)

Spreadsheet work (copy-pasted from code, to conduct analyses):

<https://docs.google.com/spreadsheets/d/1z8cZ-0Hxflcaw1d8fS9jQddN1noD1cVIIUNTUEO93ul/edit>

- What did you do?
  - First, I created a new dataframe, with one row per call. I tagged each call to one of the 11 relevant menus (10 if you don't include Farmworker), based on the last EP Name that each unique contact session ID passed through. Then, I tagged each unique call with the last Activity Name per call, creating 194 subgroups of menu & activity. Then, I aggregated these data, finding the count of number of calls for each subgroup. Meera independently followed a similar process in R (see her documentation), and we ended up with the same exact results, giving us confidence in our accuracy. Then, we used my code, copy-pasting this dataframe into Sheets and creating additional columns to tag each subgroup as Abandoned or Not Abandoned, with additional explanation. As a team, we tagged each of these subgroups, and also had someone verify each tag. For unclear cases, I developed a script where we could input menu and last activity and see the raw data, allowing us to tag every single subgroup confidently. Then, we had every subgroup tagged, and I did some additional work in sheets along with the rest of our team to calculate total % of abandonment, as well as % by menu – this work was primarily done by my groupmates and can be read in more detail in their documentation.
- How does it help the project?
  - Our objective was to find % abandonment in total, as well as per menu. This allows us to see how many callers are abandoning their calls, as well as analyzing this per

menu. This allows LegalAid to determine which of their menus are working as intended, or which ones may have confusing structure / verbiage, causing callers to abandon. Additionally, our work can help power future analyses such as successful versus unsuccessful calls by providing a baseline dataset of abandoned calls.

- Issues faced (if any)
  - Some calls never passed through the 11 relevant menus, or had no activity name
- Attempts to resolve issues (if any)
  - I created scripts to see these raw call journeys in the data.
- Issues resolved (if any)
  - We found that these were all either categorized as Intake Outdial, or had no activity name, which we determined must mean they were all outbound calls from LegalAid, meaning they could not be abandoned and thus should be excluded from our work.
- Next steps
  - We are all done! In one week, we essentially started from scratch and accomplished our objective.
- References (Mention if you built up on someone else's work)
  - Krish's CAR data import

#### Presentation 4

- Links to presentation(s) and code(s) on GitHub
  - Presentation:
    - [https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Presentations/Verbiage%20analysis/MenuAbandonment\\_18Nov\\_Adam.pdf](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Verbiage%20analysis/MenuAbandonment_18Nov_Adam.pdf)
  - Code:
    - [https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Code/Verbiage%20analysis/Call\\_Abandonment\\_18Nov\\_Adam.ipynb](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/Call_Abandonment_18Nov_Adam.ipynb)

- What did you do?
  - I identified call abandonment for the Employment, Immigration, and HIV Menu calls. This included first filtering to only calls that at some point crossed these menus, setting rules for the most basic cases of abandonment versus not abandonment as well as a column for a description of why each case was abandoned versus not abandoned, then examining the remaining calls and iteratively defining the more complex cases as either abandoned or not. I repeated this process until every call which crossed one of these 3 menus at some point was classified as abandoned, not abandoned, or not applicable (ended in a non-relevant menu screen), and represented this data in a new data frame with one row for each call, with columns for Contact Session ID, Abandoned/Not Abandoned, and Reason (description of why abandoned or not). I then created some visualizations and queries to showcase my results, segmented by menu.
- How does it help the project?
  - This works towards our group's goal of identifying all calls as either abandoned or not abandoned. In doing so, we will power additional analyses, to identify where abandoned calls are concentrated by menu, time, etc. This builds towards a better phone system.
- Issues faced (if any)
  - N/A
- Attempts to resolve issues (if any)
  - N/A
- Issues resolved (if any)
  - N/A
- Next steps
  - We have now gone through all major menus and identified calls as abandoned versus not abandoned. I think we could aggregate all of our work to provide helpful,

standardized figures and baseline information for future work that others could iterate upon.

- o Implement any feedback
- o Develop a robust plan to prevent double counting!
- References (Mention if you built up on someone else's work)
  - O Krish's CAR data import

### Presentation 3

- Links to presentation(s) and code(s) on GitHub
  - o Presentation:
    - o [https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Presentations/Verbiage%20analysis/CallAbandonment\\_05Nov\\_Adam.pdf](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Verbiage%20analysis/CallAbandonment_05Nov_Adam.pdf)
  - o Code:  
[https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Code/Verbiage%20analysis/CustomerLeftScenariosEmployment\\_05Nov\\_Adam.ipynb](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/CustomerLeftScenariosEmployment_05Nov_Adam.ipynb)
- What did you do?
  - o I identified call abandonment for the Employment Menu Calls. This included first filtering to only calls that at some point crossed the Employment Menu, setting rules for the most basic cases of abandonment versus not abandonment as well as a column for a description of why each case was abandoned versus not abandoned, then examining the remaining calls and iteratively defining the more complex cases as either abandoned or not. I repeated this process until every call which crossed the Employment Menu at some point was classified as abandoned, not abandoned, or not applicable (ended in a non-employment menu screen), and represented this data in a new data frame with one row for each call, with columns for Contact

Session ID, Abandoned/Not Abandoned, and Reason (description of why abandoned or not). I then created some visualizations and queries to showcase my results.

- How does it help the project?
  - This works towards our groups goal of identifying all calls as either abandoned or not abandoned. In doing so, we will power additional analyses, to identify where abandoned calls are concentrated by menu, time, etc. This builds towards a better phone system.
- Issues faced (if any)
  - At first, we all had different amounts of data. Also, we have the possible issue of overlapping calls, for example if a caller at some point enters both the employment menu and another menu.
- Attempts to resolve issues (if any)
  - We set our data import processes to be the same and standardized to 90,940 calls after March 2025. To fix the overlapping calls issue, I created plenty of rules to identify calls that did not end in Employment and did not count these as abandoned or not abandoned. If this process is perfect, we will have no double counting.
- Issues resolved (if any)
  - Yes, described above
- Next steps
  - Make a plan to identify all calls as either abandoned or not abandoned together
    - I was thinking this might actually be easier as a macro-project, rather than menu-by-menu to create simpler code, but uncertain.
  - Implement any feedback
  - Develop a robust plan to prevent double counting!
- References (Mention if you built up on someone else's work)
  - Krish's CAR data import

## Presentation 2

- Links to presentation(s) and code(s) on GitHub
  - o Presentation:

[https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Presentations/Verbiage%20analysis/CallJourneyVerbiage\\_21Oct\\_Adam.pdf](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Verbiage%20analysis/CallJourneyVerbiage_21Oct_Adam.pdf)

- o Code:

[https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Code/Verbiage%20analysis/CallMenuVerbiage\\_21Oct\\_Adam.ipynb](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/CallMenuVerbiage_21Oct_Adam.ipynb)

- What did you do?
  - o I conducted two data analyses to test my hypotheses about inefficiencies in the call menu verbiage/structure. First, I conducted an analysis to determine the final queue destination of calls; Specifically, I aimed to calculate the number and % of calls which ended in the clinic voicemail transfer, as opposed to a different destination. I found that about 24% of calls ended at the clinic voicemail transfer, which does not provide sufficient evidence that the clinic voicemail transfer (or option 4) should be put before option 3. HOWEVER, this does not really test my initial question, but I unfortunately don't know if it is feasible with the data, so this may be a dead end.
  - o Next, I conducted an analysis related to how many callers used the complaint/compliment phone line and did not back out to the main menu. I found that 15 calls reached an agent from the call/complaint menu, meaning that resources are being used here. Recommendation: Consider removing the option to use the phone line to leave a compliment or complaint, either by just using the website or using a voicemail.
- How does it help the project?
  - o The first analysis did not yield very helpful results. From the second analysis, we see that phone agent resources are being used for compliments and complaints, and

this could be used to bypass the queue system and reach the front desk. Thus, I would tentatively recommend that removing this option or converting it to a voicemail box would increase efficiency and clear up resources.

- Issues faced (if any)
  - My first attempt at the compliment/complaint analysis was flawed. After looking at the output dataset, I saw that most of the people who selected compliment/complaint did so on accident and thus backed out of the queue.
- Attempts to resolve issues (if any)
  - I added an additional filter to filter these values out! Now, when I analyze the output data, it is logical and only shows people who actually used the compliment/complaint without backing out.
- Issues resolved (if any) ◦ Yes, described above
- Next steps
  - Determine / quantify how much time resources, etc. could be saved by implementing my recommendation about removing the compliment/complaint phone line
  - Brainstorm alternatives to increase access for those who don't have internet access
  - Use above to design and make a final, clear, structured and data-driven recommendation
  - Open to additional ideas as well!
- References (Mention if you built up on someone else's work) ◦ Krish's CAR data import

## Presentation 1

- Links to presentation(s) and code(s) on GitHub
  - [https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Code/Verbiage%20analysis/CallMenuVerbiage\\_13Oct\\_Adam.ipynb](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Verbiage%20analysis/CallMenuVerbiage_13Oct_Adam.ipynb)
  - [https://github.com/arvindkrishna87/STAT390\\_LegalAid\\_Fall2025/blob/main/Presenations/Verbiage%20analysis/CallJourneyVerbiage\\_13Oct\\_Adam.pdf](https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presenations/Verbiage%20analysis/CallJourneyVerbiage_13Oct_Adam.pdf)
- What did you do?

- o I started with a qualitative comprehensive analysis of all of the verbiage within the phone menu. With each prompt, I imagined that I was a caller with a specific issue and thought of spots where I may get mixed up, be between two options, or generally end up in the queue which is not best suited for me, ultimately decreasing LegalAid's efficiency and ability to serve the most people, as well as potentially causing confusion regarding what somebody's reason for calling is. This included going back on my first presentation and removing one recommendation which was

not very good, and coming up with 4 total recommendations, each of which I think is very solid and will improve LegalAid's menu comprehensibility.

- How does it help the project?
  - o More efficient and less confusing customer-facing verbiage will allow for a better user experience and a more streamlined journey from calling to getting into the right queue, as well as potentially decreasing the load on LegalAid's phone system. Based on Cynthia's feedback, I can understand if these are all 4 interesting questions to explore, or if I should be more selective, allowing me to continue with data analysis this week to test my theories.
- Issues faced (if any):
  - o At first, I was having trouble with efficiently accessing the data, but Krish's Notebook helped me with this. I also had some trouble understanding how the queues and phone journeys connected.
- Attempts to resolve issues (if any)
  - o I used and ran the All Calls Data import to consolidate the different months into one data file that I will use to power my analyses next presentation! Additionally, I found the flowchart of call journeys to understand how the lines connect.
- Issues resolved (if any)
  - o Solved trouble with efficiently accessing data, understand flowchart



- Next steps   o Now that I have an array of qualitative verbiage analysis, I will bring in my quantitative testing to develop and execute a plan to determine if I am correct and make actionable and data-backed recommendations.
- References (Mention if you built up on someone else's work)